```
function [a,emag,i,omega,w,nu] = calcOrbElems(mu,rvec,vvec)

% function to calculate the orbital elements given a pos and vel vec
        %input: mu, position vector, velocity vector
        %output: classical orbital elements

rmag = norm(rvec);
vmag = norm(vvec);

% calculate the orbital elements for this satellite

hvec = cross(rvec,vvec);
hmag = norm(hvec);

% eccentricity
evec = ((cross(vvec,hvec) - (mu*(rvec))/rmag))*(1/mu);
emag = norm(evec);

% semi major axis
p = hmag^2/mu;
a = p/(1-emag^2);

% inclination

kvec = [0;0;1];
i = acosd((dot(hvec,kvec))/hmag);

% argument of periapsis

invec = cross(kvec,hvec)/(norm(cross(kvec,hvec)));
wcalc = acosd(dot(evec,invec)/emag);

% quadrant check:
% w_check = dot(evec, kvec);
%
% if w_check < 0
%     w = 360 - wcalc;
% elseif w_check > 0
%     w = wcalc;
% else
%     disp("undefined")
% end

if dot(cross(invec, evec), hvec) < 0
    w = 360 - wcalc;
else
    w = wcalc;
end

% RAAN
```

```
nvec = cross([0,0,1],hvec);
n = norm(nvec);

omega = acosd(nvec(1)/n);

if nvec(2) <= 0
    omega = 360 - omega;
end

% true anomaly

nucalc = acosd(dot(evec,rvec)/(emag*rmag));
nu_check = dot(rvec,vvec);

% checking quadrant for true anomaly
if nu_check < 0
    nu = 360 - nucalc;
elseif nu_check >= 0
    nu = nucalc;
else
    disp("undefined")
end


end
```